DP Class - 3

Special class

Love Babbar • Sept 18, 2024

→ Knapsack Problem → 0/1 → DP

→ Fractional → greedy

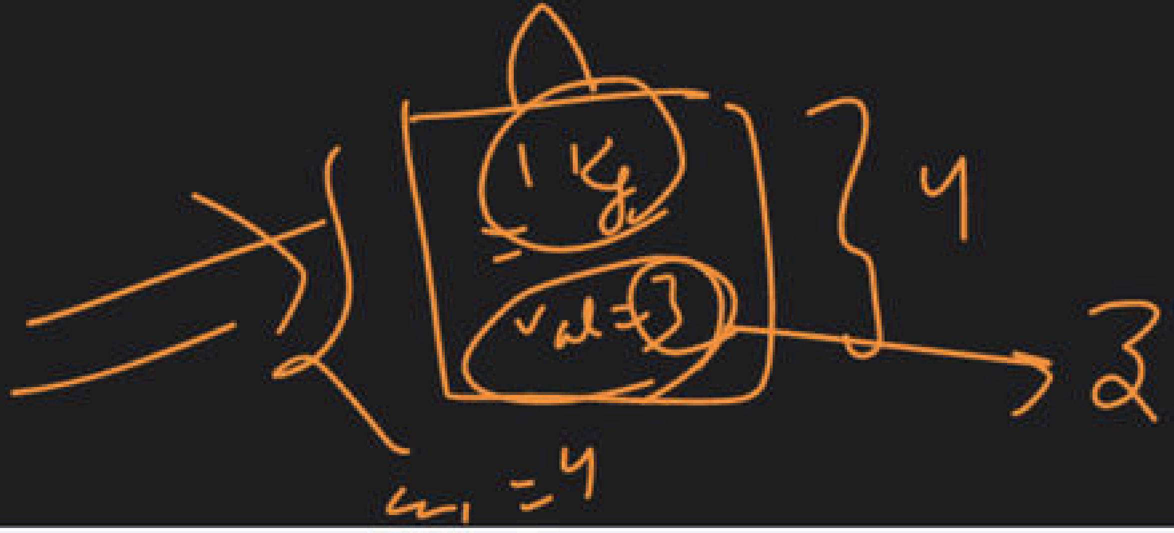
items → item1 , item2 , item3

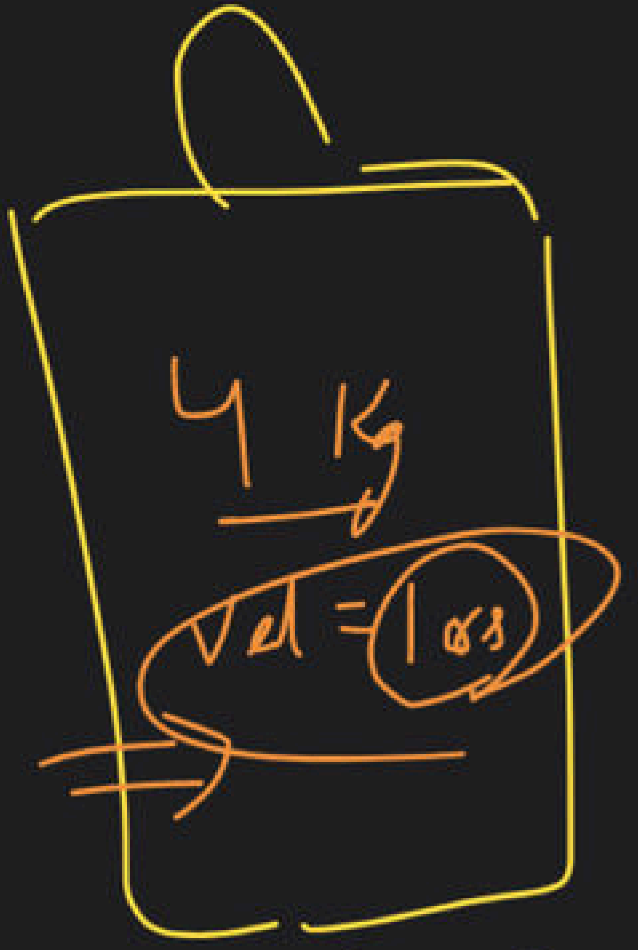
wt → { 4 kg , 5 Kg , 1 Kg }

Val[ ] → { 1 , 2 , 3 }.

Subsequence

inc / excl





Capacity → 4 Kg

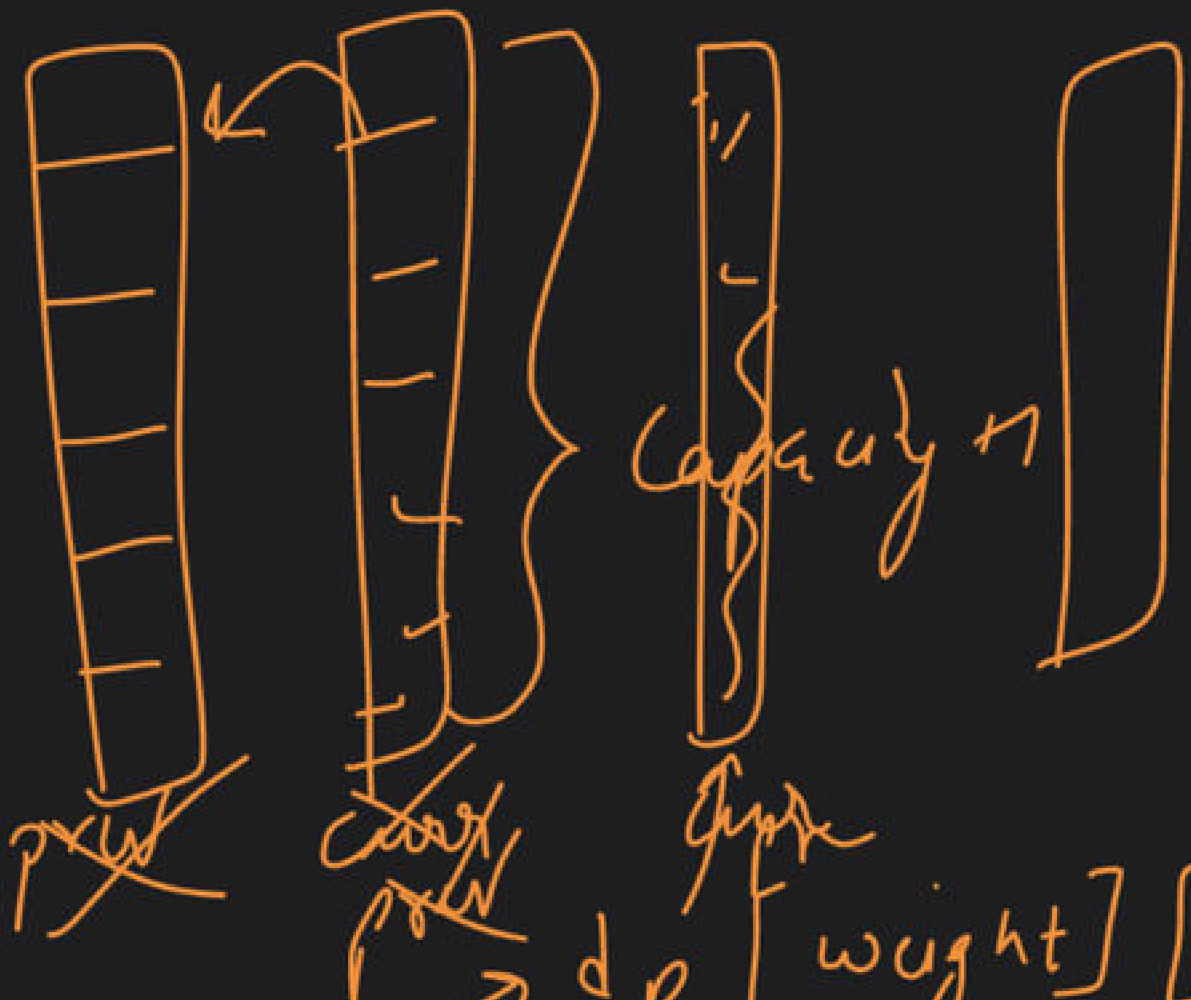
S.O

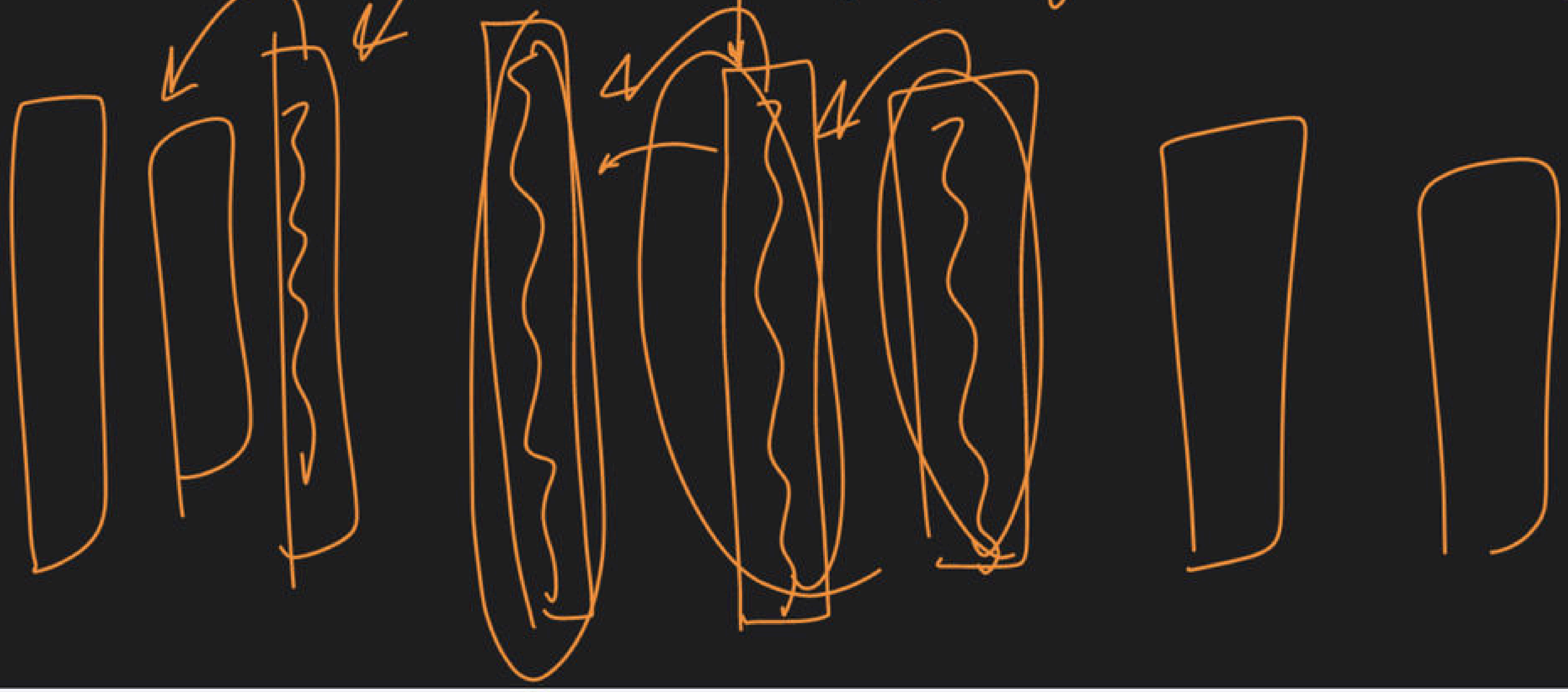
dependency

dp[weight-1][index]

dp[weight][index-1]

dp[weight - wt[index]][index-1]

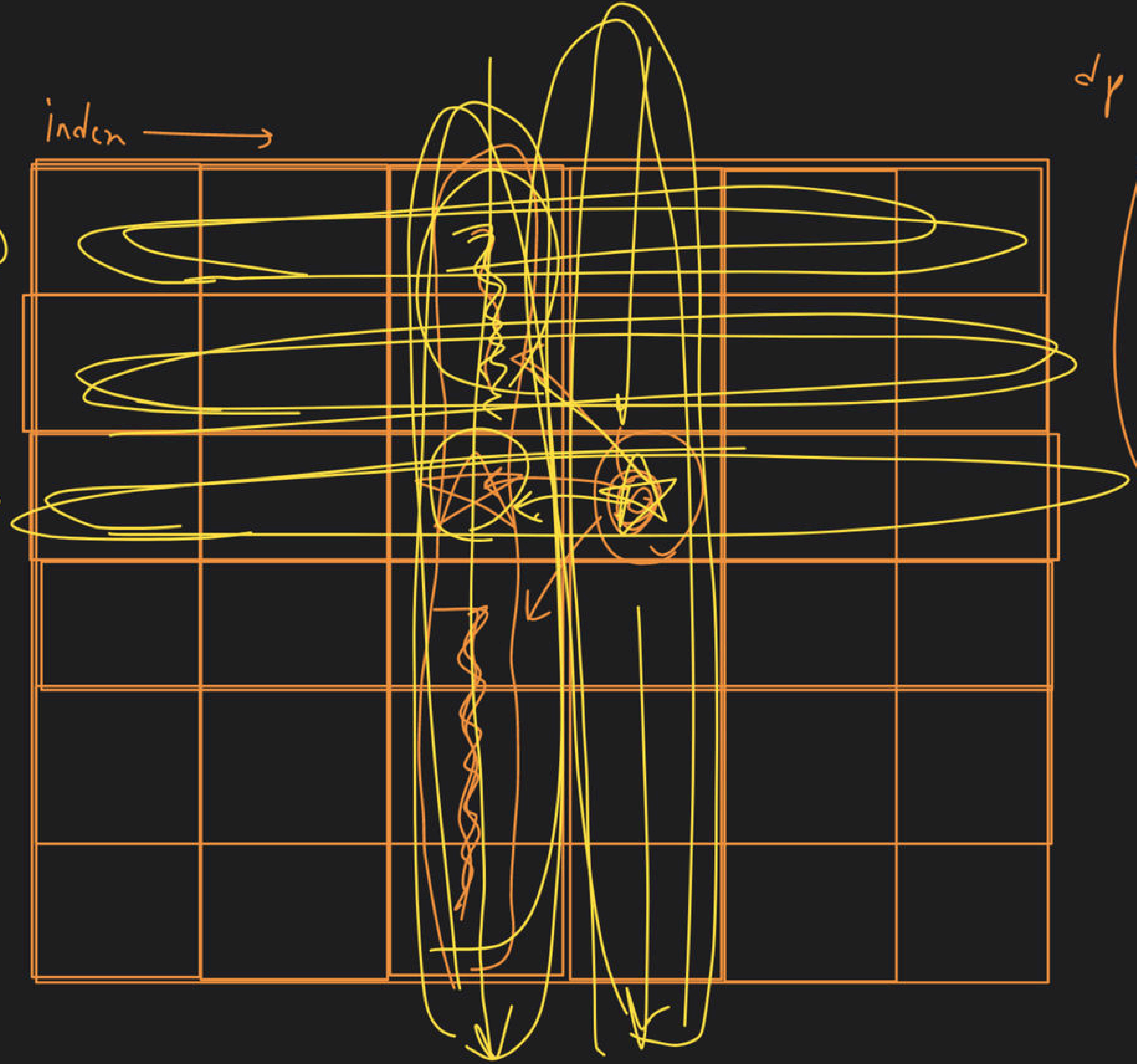




index →

capacity

0



dp[wt][in]

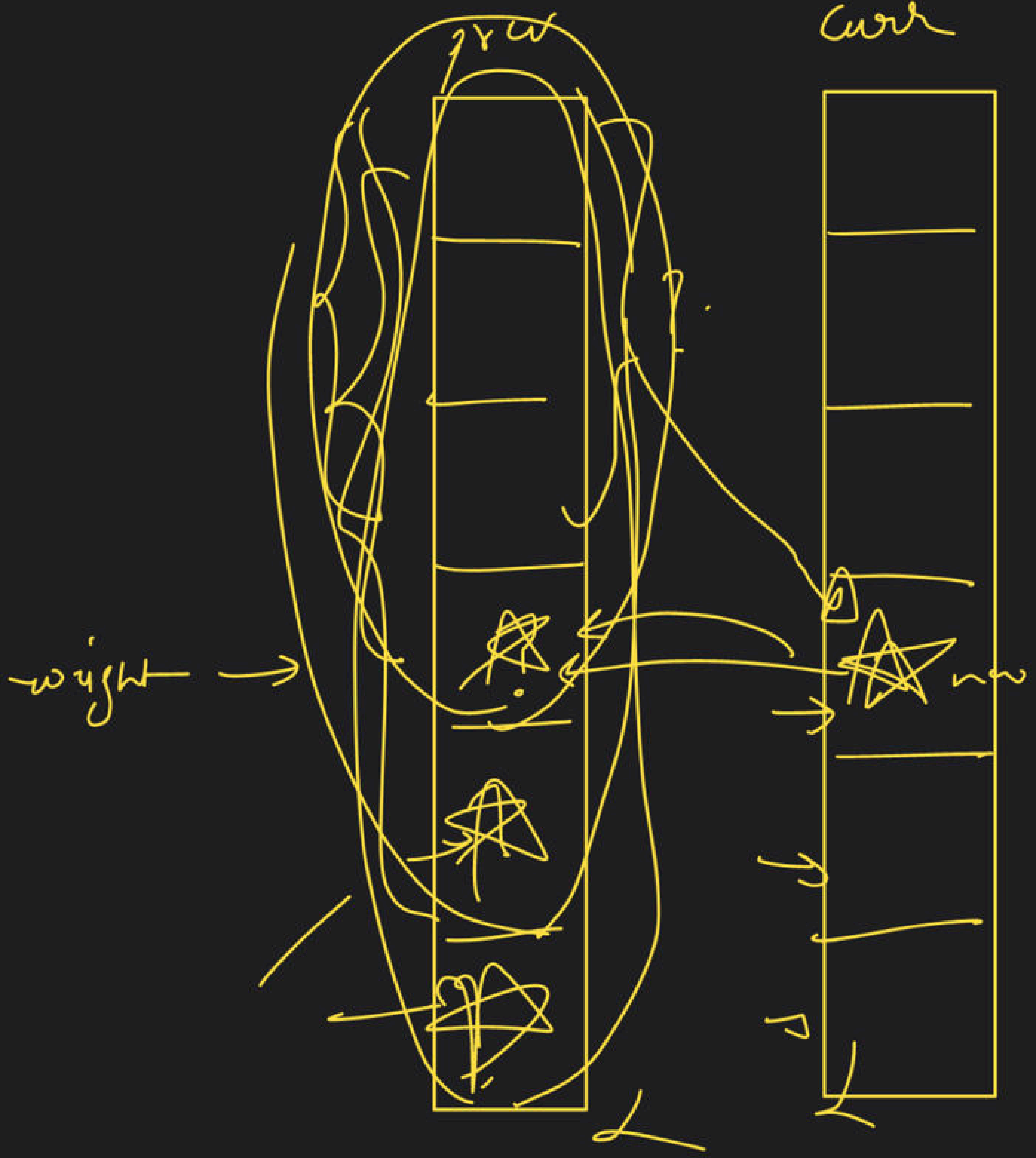
dp[wt][in-1]

dp[wt - wt][in-1]

<= wt

for(wt → 0 → capacity)

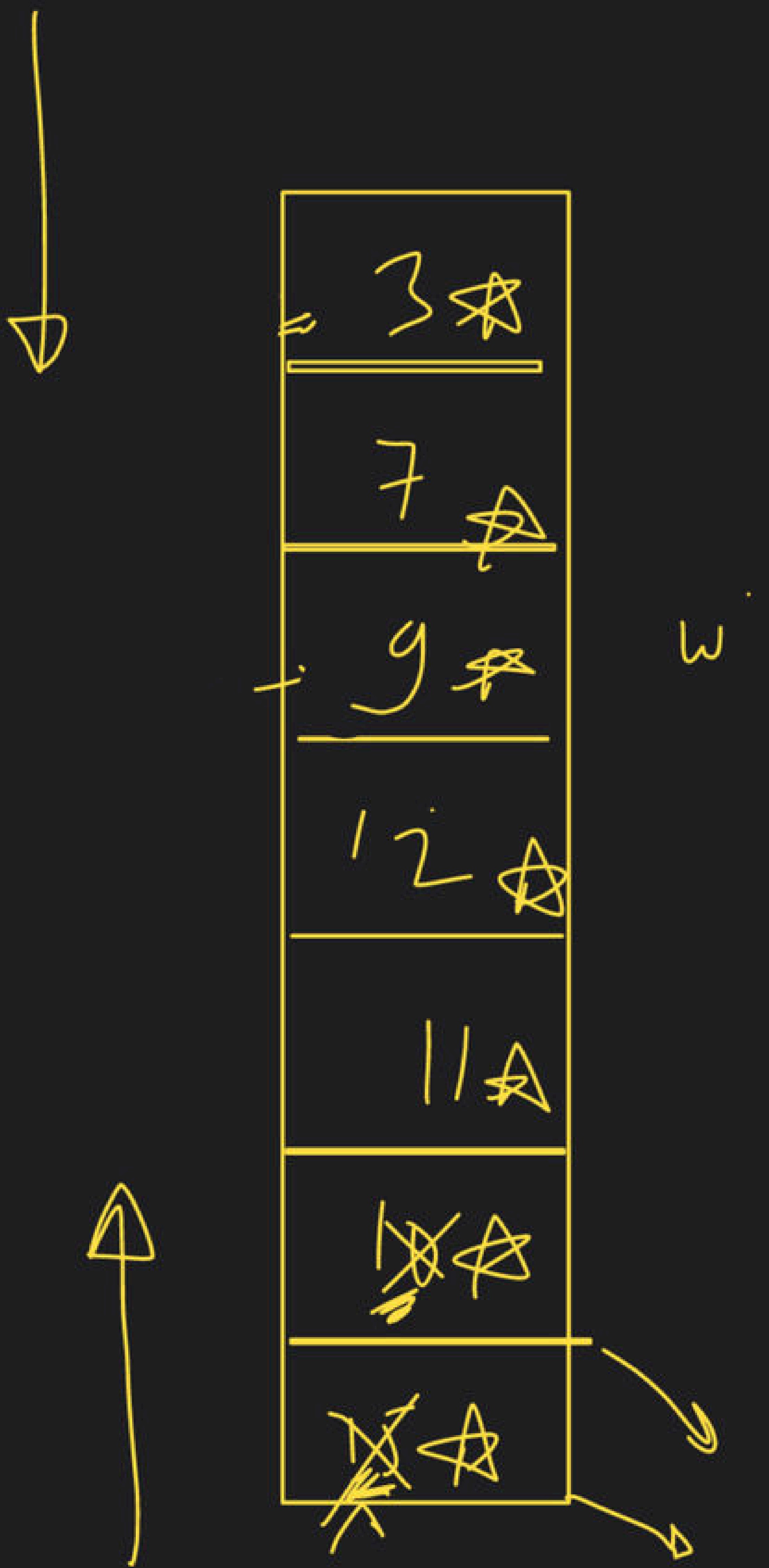
for(1 → n-1)



curr[weight]

→ prev[weight]

→ prev[weight - wt[i]]



~~for~~

for (index = 1 → (n-1))

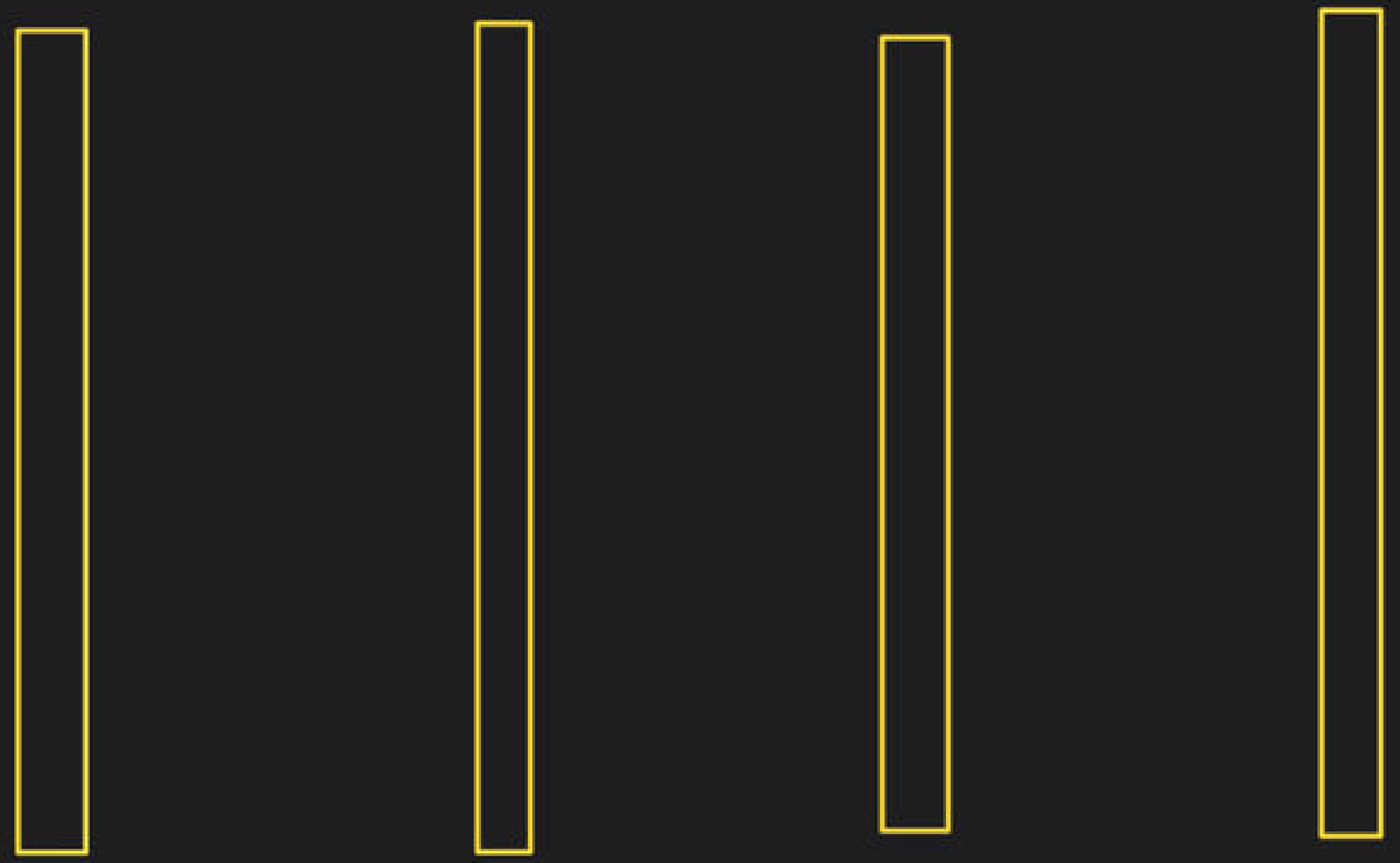
for (weight ~~capacity~~ → 0 → W)

arr[weight]

-> Painting fence problem

R G B R

R R G G



D_1 D_2 D_3 D_4

$\underbrace{\qquad\qquad\qquad\qquad}_{n}$

total no of ways to paint the fence

K

colors -> 3

R G B

R ✓ RG ✓

G ✓ RR

RR ✓ B ✓ BB

GG ✓ GG

BB RRR

RRR

RRRR

$K = 3 \rightarrow RGB$ $\quad n=1 \rightarrow 3$ $\quad n=2 \quad 9$

	$n=1$	$n=2$	$n=3$	$n=4$
same	R G B	RR GG BB	RGG RBB GRR GBB BRR BGG $f(n-2) * (K-1)$	$f(n-2) * (K-1)$ RR, GG, BB $\rightarrow K$
diff	ans = 3	RG RB GR GB BR BG	RR $(K-1)$ GG $(K-1)$ BB RG RB GB GR BR BG $f(n-1) * (K-1)$	$f(n-1) * (K-1)$ R, G, B $f(n-2) * (K-1) + f(n-1) * (K-1)$ $(K-1) * (f(n-1) + f(n-2))$

same

x x

4 A

4 4

B D

K

$K + (K) * (K-1)$

x x

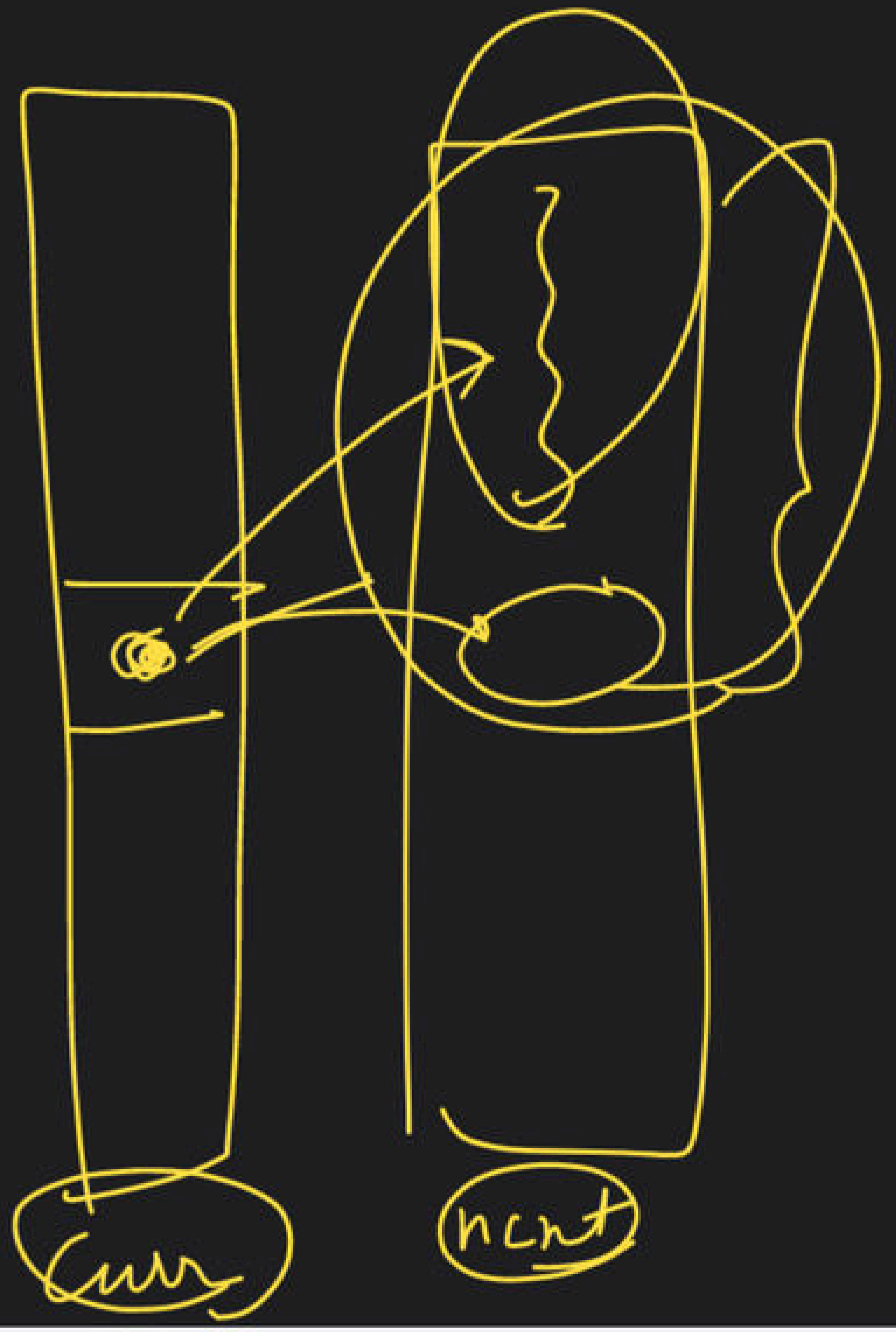
B G B	2

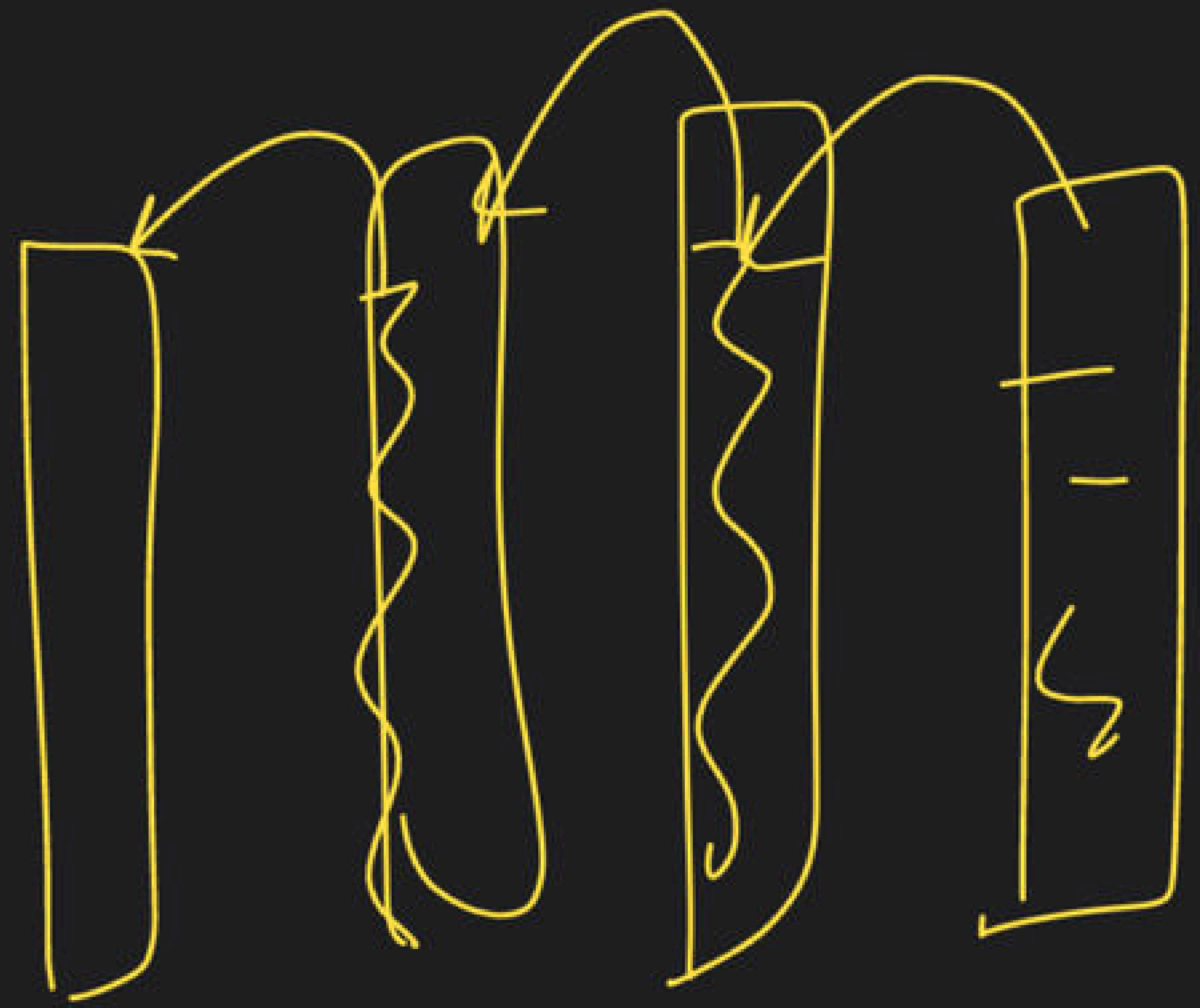
$3 \times 2 \rightarrow 6$

$K * (K-1)$

$$dp[t][i] \rightarrow dp[t][i+1]$$

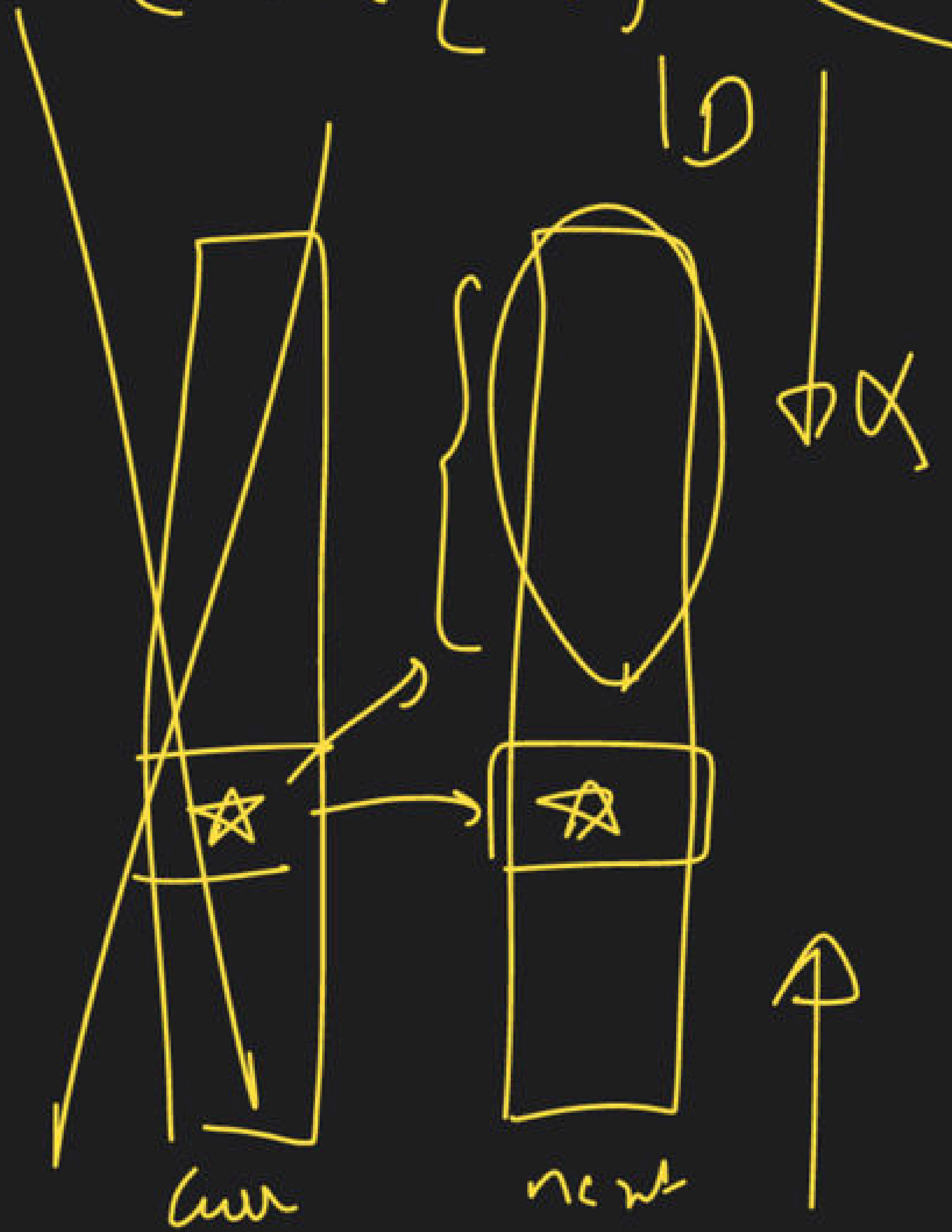
$$dp[t][i] \rightarrow dp[t - nums[i]][i+1]$$





$curr[t] \rightarrow next[t]$

$curr[t] \rightarrow next[t - nums[i]]$



i/p

→ n → no. of dice

K → faces

target → sum

total possible ways

o/p

$n = 2$, $K = 6$

$t = 9$

$\{1, 1\}$	$\{2, 1\}$	$\{3, 1\}$	$\{4, 1\}$	$\{5, 1\}$	$\{6, 1\}$
$\{1, 2\}$	$\{2, 2\}$	$\{3, 2\}$	$\{4, 2\}$	$\{5, 2\}$	$\{6, 2\}$
$\{1, 3\}$	$\{2, 3\}$	$\{3, 3\}$	$\{4, 3\}$	$\{5, 3\}$	$\{6, 3\}$
$\{1, 4\}$	$\{2, 4\}$	$\{3, 4\}$	$\{4, 4\}$	$\{5, 4\}$	$\{6, 4\}$
$\{1, 5\}$	$\{2, 5\}$	$\{3, 5\}$	$\{4, 5\}$	$\{5, 5\}$	$\{6, 5\}$
$\{1, 6\}$	$\{2, 6\}$	$\{3, 6\}$	$\{4, 6\}$	$\{5, 6\}$	$\{6, 6\}$

10 discs

{ 1 — — — — — — — — — — 3

{ 2 — — — — —

{ 3 — — —

{ 4 — — — — — Rec

{ 5 —

{ 6 — — — — —

$$dp[N][t] \longrightarrow dp[\underline{N-1}][\underline{t-i}]$$

$N \times N$